

Karthik Viswanathan

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Education

University of Amsterdam

Amsterdam, Netherlands

PhD in Physics, Expected: December 31, 2025

2021–2025

Thesis: From Language Models to Cosmic Structures: A Geometric Perspective

Advisor: Dr. Jan Pieter van der Schaar

University of Amsterdam

Amsterdam, Netherlands

MSc in Physics and Astronomy (Theoretical Physics), cum laude

2019–2021

Thesis: Exploring the Spectral Theory/Topological Strings duality

Advisor: Dr. Marcel Vonk

GPA: 8.7/10

Indian Institute of Technology Madras

Chennai, India

B.Tech in Engineering Physics

2013–2017

Thesis: Real Space Renormalization and Applications to Machine Learning

Advisor: Dr. Ashwin Joy

GPA: 8.29/10

Research Experience

Incentivizing Externalization of Reasoning in LLMs via Early Exit Mechanisms.

Geodesic Research, Cambridge AI Safety Hub

Research done as part of Mentorship for Alignment Research Students (MARS 3.0)

Resources: code

Probing Geometry of Next Token Prediction Using Cumulant Expansion of Softmax Entropy

Karthik Viswanathan, Sang Eon Park

ICML 2025: Workshop on High-dimensional Learning Dynamics

Resources: paper, code

Persistent Topological Features in Large Language Models

Yuri Gardinazzi*, Karthik Viswanathan*, Giada Panerai, Alessio Ansuini, Alberto Cazzaniga, Matteo Biagetti

Proceedings of the 42nd International Conference on Machine Learning

*Equal contribution

Resources: paper, code

The Geometry of Tokens in Internal Representations of Large Language Models

Karthik Viswanathan, Yuri Gardinazzi, Giada Panerai, Alberto Cazzaniga, Matteo Biagetti

Under review

Resources: paper, code

Cosmology with persistent homology: A Fisher forecast

Jacky Yip, Matteo Biagetti, Alex Cole, Karthik Viswanathan, Gary Shiu

Journal of Cosmology and Astroparticle Physics (JCAP), Vol. 2024, Issue 09

Resources: paper

Workshop: Interpretability in LLMs using Geometric and Statistical methods

Karthik Viswanathan, Jan Pieter van der Schaar

Organized a two-day workshop with around 40 participants from Amsterdam and AREA Science Park (Trieste).

Professional Experience

Goldman Sachs

Surveillance Analyst

Developed ML models for anomaly detection in financial data

Manager: Prof. Howard Karloff

Bangalore, India

2017–2019

Goldman Sachs

Summer Intern

Implemented fast Personalized PageRank in a MapReduce framework

Mentor: Dr. Koushik Balasubramanian

Bangalore, India

2016

Awards & Distinctions

Sander Bais Prize for Academic Merit

Institute for Theoretical Physics Amsterdam

For exceptional academic performance in the master's program

Amsterdam, Netherlands

2020

Honorable Mention

ACM ICPC World Finals

Represented India in the international collegiate programming contest.

Marrakech, Morocco

2015

Participant

International Olympiad in Informatics Training Camp

Selected after ranking in top 25 in the Indian National Olympiad in Informatics

Bangalore, India

2013

Organization & Mentoring

AREA Science Park, Trieste

Master Student Supervisor

Student: Giada Panerai

Thesis: Zig-Zag Persistence in Neural Networks Representations

2024

University of Amsterdam

Master Student Mentor

Student: Sibilla Bouche

Thesis: The persistence of non-Gaussian features: a Neural Ratio Estimation approach

2023

University of Amsterdam

Master Student Supervisor

Student: Sliem el Ela

Thesis: From Primordials to Persistence: A Dual Exploration of Multiparameter Topology and Cosmic Origins

2023

University of Amsterdam

Journal Club Organizer

Cosmology Journal Club

2022

Technical Skills

Programming: PyTorch, TensorFlow, Hadoop, MapReduce, Mathematica

HPC Resources: LUMI consortium, Snellius supercomputer

Teaching

University of Amsterdam

Teaching Assistant

Courses: Topological Data Analysis: a Physics Perspective, Advanced Cosmology, ML for Physics and Astronomy, QFT in Curved Spacetimes, General Relativity

MSc Physics Program

2020–2024

Research Visits

2024: AREA Science Park, Trieste, Italy

September 16 – October 4

2024: AREA Science Park, Trieste, Italy

February 15 – April 27

References

Dr. Jan Pieter van der Schaar

University of Amsterdam

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Dr. Matteo Biagetti

AREA Science Park, Trieste

matteo.biagetti@areasciencepark.it

Dr. Magnus Botnan

Vrije Universiteit, Amsterdam

M.B.Botnan@vu.nl

Dr. Alberto Cazzaniga

AREA Science Park, Trieste

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